Verovio

Music engraving in Typst

github.com/bernsteining/verovio · typst.app/universe/package/verovio

Contents

Quick Start 1
API Reference 1
`render-music` 1
`music-page-count` 1
Verovio Options 2
Music Fonts 3
Supported Input Formats 4
Exporting from notation software 4
ABC Notation 4
MusicXML 5
MEI (Music Encoding Initiative) 6
Humdrum 7
Volpiano 8
CMME 8

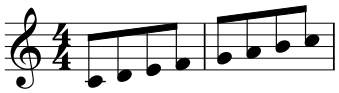
Verovio — Music Engraving Plugin for Typst

Verovio is a music notation engraving library. This Typst plugin wraps Verovio as a WebAssembly module, rendering music from multiple input formats directly into SVG embedded in your document.

Quick Start

Render inline ABC notation:

```typst
#import "@local/verovio:0.1.0": render-music
#render-music("X:1\nM:4/4\nK:C\nCDEF|GABc|")
```

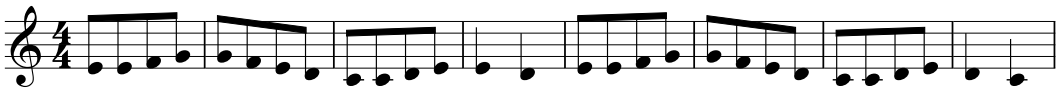


Or define a show rule to render ABC code blocks automatically:

```typst
#import "@local/verovio:0.1.0": render-music
#show raw.where(lang: "abc"): it => render-music(it.text)

```abc
X:1
T:Ode to Joy
M:4/4
K:C
EEFG|GFED|CCDE|E2D2|
EEFG|GFED|CCDE|D2C2|
```
```

Ode to Joy



API Reference

`render-music`

```typst
#render-music(
  data,             // string: music data (ABC, MusicXML, MEI, Humdrum, Volpiano, CMME)
  options: none,    // dictionary: verovio options
  page: 1,          // int: page number to render
  ..args,           // forwarded to Typst's image() (width, height, fit, alt)
)
```

`music-page-count`

```typst
#let n = music-page-count(data, options: none)
```

Returns the number of pages for the given music data and options. Useful to loop over pages of a multi-page score:

Verovio Options

Options are passed as a Typst dictionary. They map directly to Verovio's toolkit options.

Common		
`adjustPageHeight`	`true`	Crop SVG height to content
`adjustPageWidth`	`false`	Crop SVG width to content
`scale`	`100`	Scale factor (percent)
`font`	`"Leipzig"`	Music font: Leipzig, Bravura, Gootville, Leland, Petaluma
`inputFrom`	`"auto"`	Format: auto, mei, musicxml, abc, humdrum, volpiano, cmme
`pageWidth`	`2100`	Page width (MEI units)
`pageHeight`	`2970`	Page height (MEI units)
`pageMarginTop`	`50`	Top margin
`pageMarginBottom`	`50`	Bottom margin
`pageMarginLeft`	`50`	Left margin
`pageMarginRight`	`50`	Right margin
`landscape`	`false`	Landscape orientation
`breaks`	`"auto"`	Line breaks: auto, line, encoded, none
`condense`	`"auto"`	Condense: auto, none, encoded
`transpose`	`""`	Transpose (e.g. "M2" for major second up)
`header`	`"auto"`	Header: auto, none, encoded
`footer`	`"auto"`	Footer: auto, none, encoded
Layout		
`spacingStaff`	`12`	Spacing between staves
`spacingSystem`	`12`	Spacing between systems
`spacingLinear`	`0.25`	Linear spacing factor
`spacingNonLinear`	`0.6`	Non-linear spacing factor
`unit`	`9`	Base unit size (half staff space)
`stemWidth`	`0.2`	Stem width
`barLineWidth`	`0.3`	Bar line width
`staffLineWidth`	`0.15`	Staff line width
`lyricSize`	`4.5`	Lyrics font size
`hairpinSize`	`3.0`	Hairpin height
SVG Output		
`svgViewBox`	`false`	Use viewBox instead of width/height
`svgRemoveXlink`	`false`	Use href instead of xlink:href
`svgBoundingBoxes`	`false`	Add bounding box rects (debug)
`removeIds`	`false`	Strip element IDs from SVG
`smuflTextFont`	`"embedded"`	SMuFL text font: embedded, linked, none

Full reference: https://book.verovio.org/toolkit-reference/toolkit-options.html

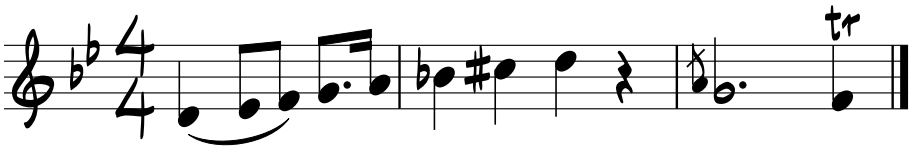
Music Fonts

Five SMuFL-compliant music fonts are available. Set the font with the `font` option:

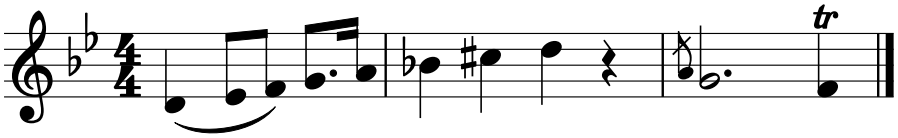
typst

```typst
#render-music(data, options: (font: "Petaluma"))
```

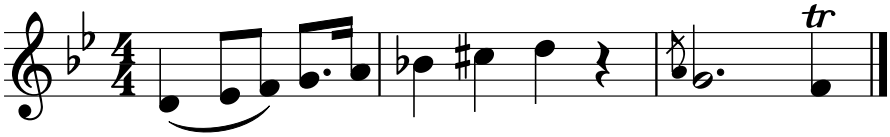
Petaluma



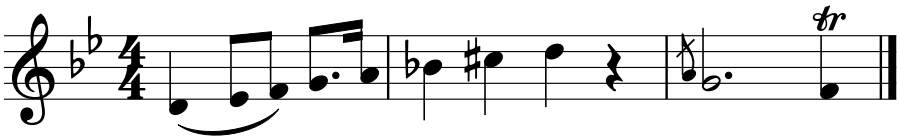
Leland



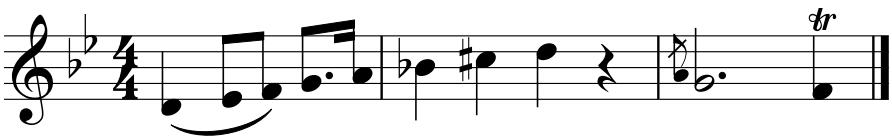
Gootville



Bravura



Leipzig



Supported Input Formats

Verovio auto-detects the input format for ABC, MusicXML, MEI, and Humdrum. For Volpiano and CMME, pass `inputFrom` explicitly.

All the files used in the examples are available in the project's Github.

Exporting from notation software

If you're working with any of these scoring software this table sums up each supported export format with their documentation linked.

Software	MusicXML	MEI	ABC	Humdrum
MuseScore	✓	✓		
Finale	✓			
Sibelius	✓			
Dorico	✓			
LilyPond	✓			
Flat.io	✓			
Noteflight	✓			
ABC tools	✓		✓	
Humdrum tools	✓	✓	✓	✓
MEI tools	✓	✓		

MusicXML is the universal interchange format — virtually all notation software can export it. For the best results, export as uncompressed MusicXML (`.musicxml` or `.xml`, not `.mxl`).

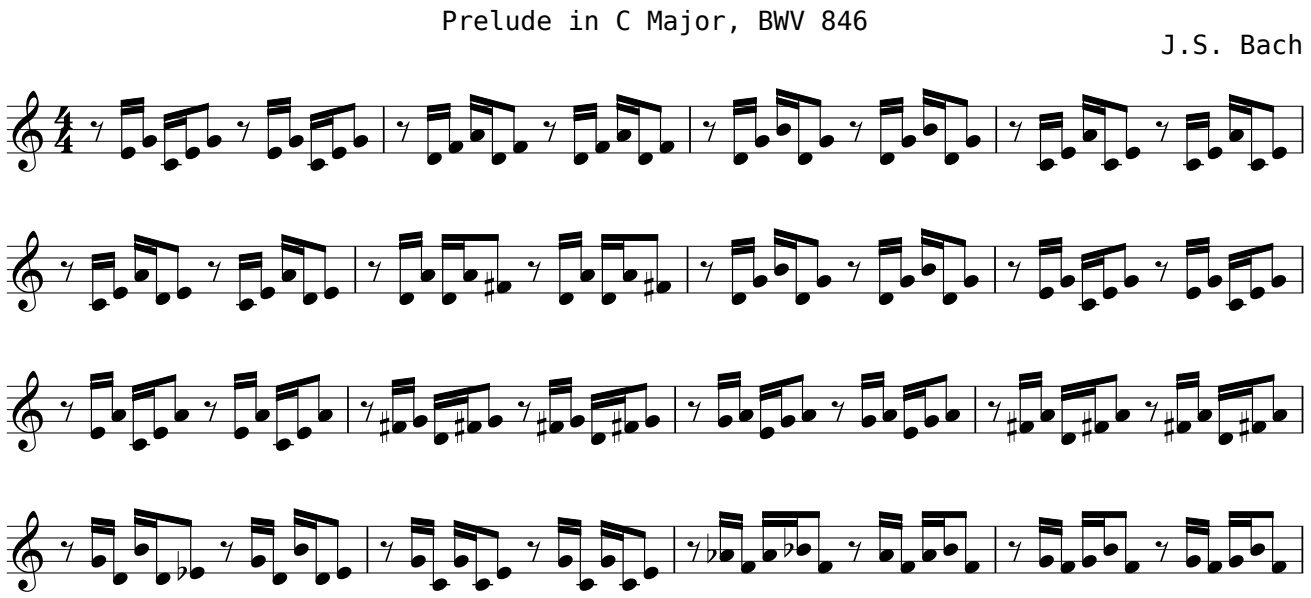
ABC Notation

ABC specification · Download ABC files

ABC is a compact text-based format popular for folk and classical melodies. Chord symbols are supported.

typst

```
#render-music(read("bach-prelude-cmaj.abc"))
```



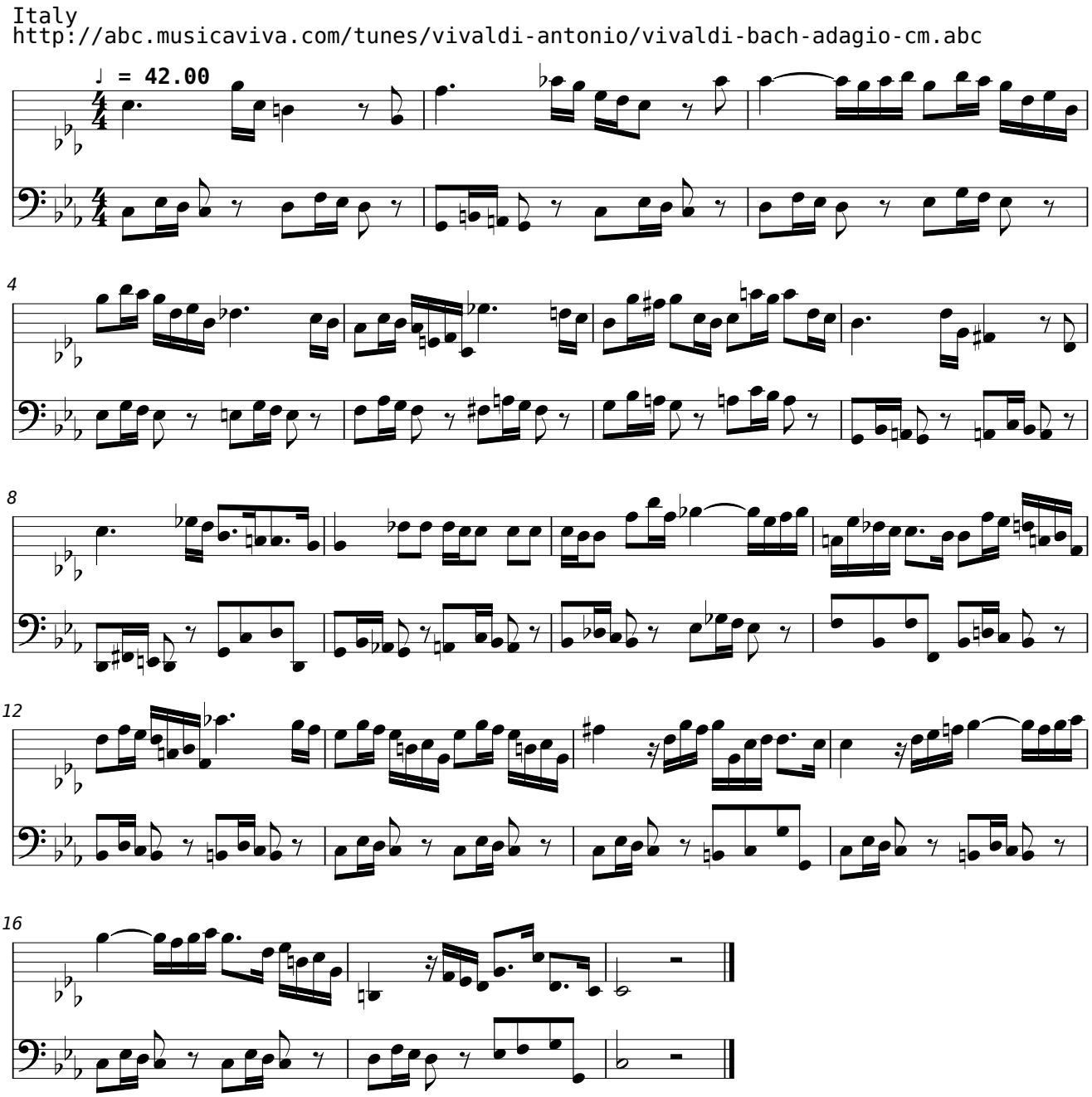
MusicXML

MusicXML 4.0 specification · Download MusicXML files

MusicXML is the standard interchange format for notation software. It supports grand staff, multiple voices, dynamics, and full score layout.

typst

```
#render-music(read("adagio.xml"))
```



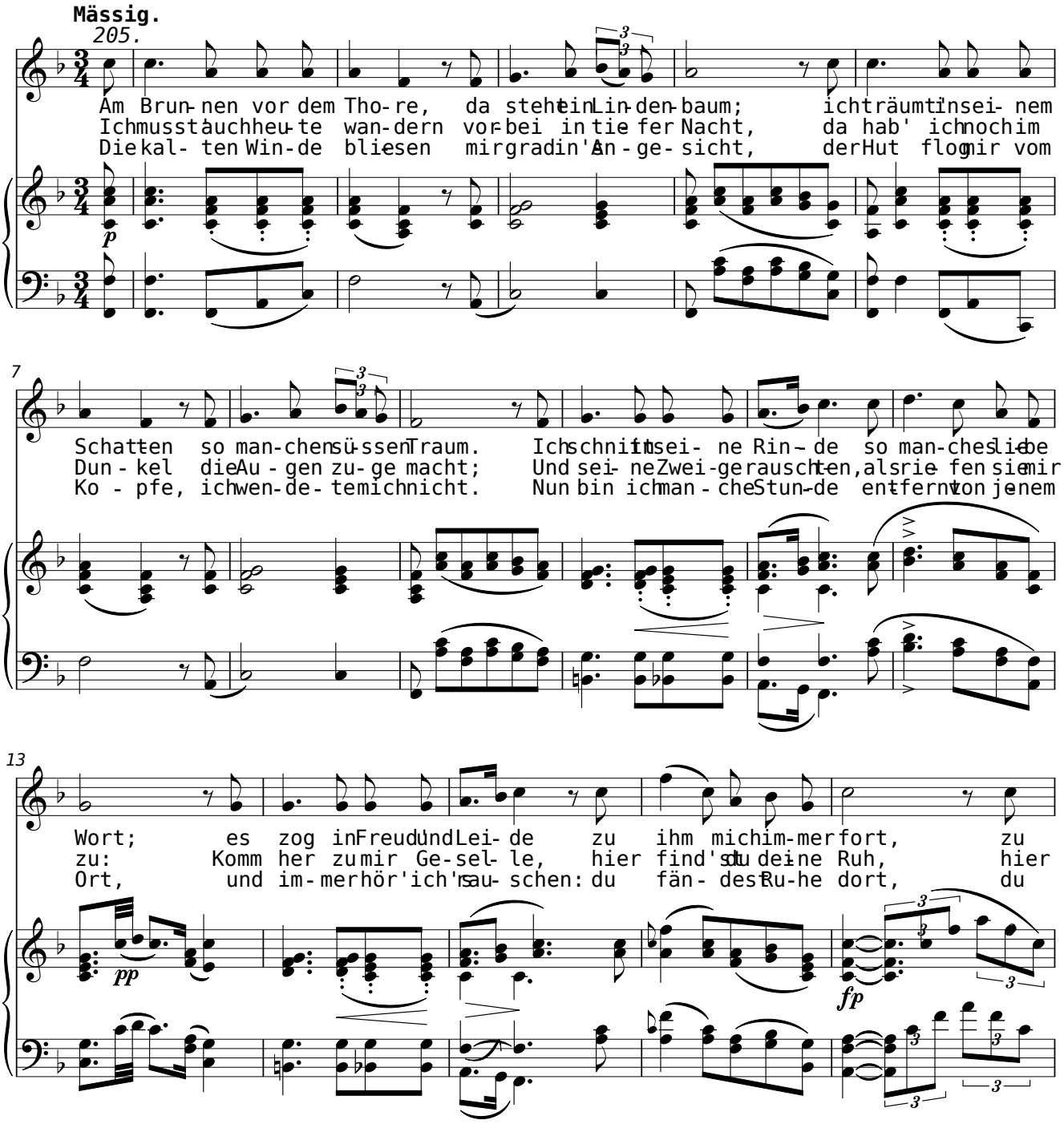
MEI (Music Encoding Initiative)

MEI Guidelines · Sample MEI encodings

MEI is a rich XML-based format used in musicology, supporting lyrics, polyphonic textures, fermatas, and detailed editorial markup.

typst

```
#render-music(read("schubert.mei"))
```



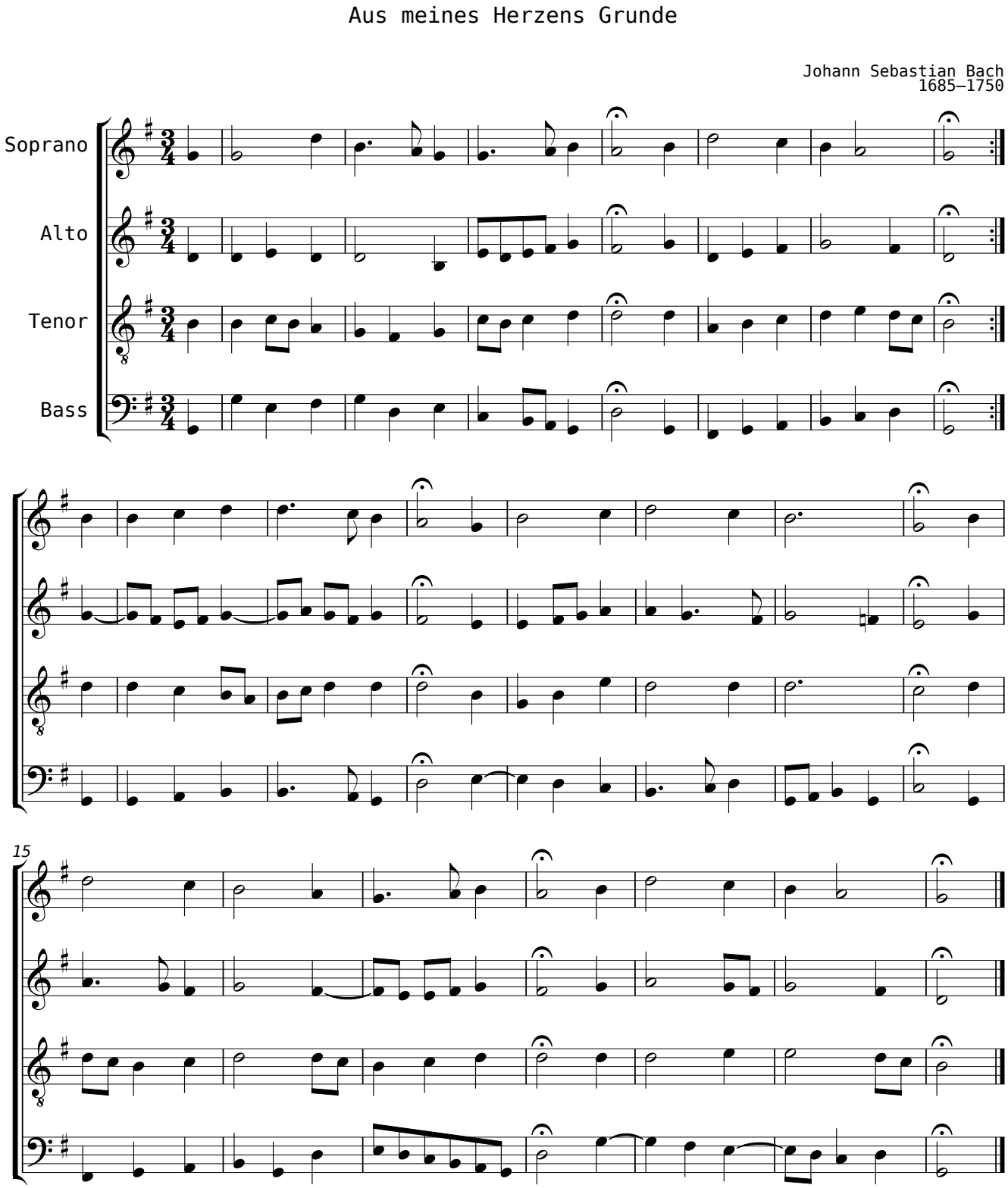
Humdrum

Humdrum User Guide · Download kern files

Humdrum uses a tab-separated spine structure with **kern encoding for pitches and durations. Widely used in computational musicology.

typst

```
#render-music(read("sample-humdrum.krn"))
```



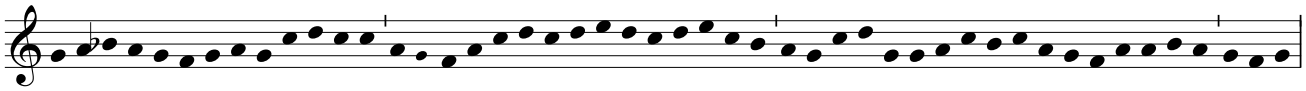
Volpiano

Volpiano specification · CANTUS database

Volpiano is a text encoding for medieval chant notation, used by the CANTUS database. Here is *Veni Creator Spiritus*, the famous Pentecost hymn. Requires `inputFrom: "volpiano"`.

typst

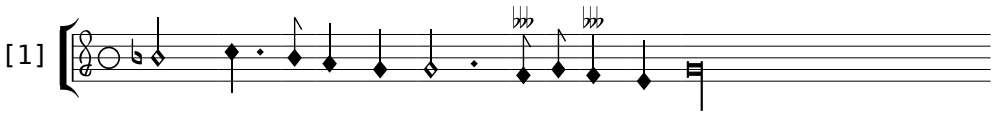
```
#render-music("1---g--hij---hgf--g--hg---k--lk--k7---hG--f---h--k--lk---l--m--l---k--lm---kj7--hg--kl---g--gh--k---
jk---h---gf--h--hjh7---g--f--g7---3", options: (inputFrom: "volpiano"))
```



CMME

CMME project · Download CMME files

CMME is an XML format for mensural notation (medieval and Renaissance music). Requires `inputFrom: "cmme"`.

typst

```
#render-music(read("cmme.xml"), options: (inputFrom: "cmme"))
```